Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry–institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2025-26 (ODD)**	**Course Name**	**Web Technology Lab**
Semester	**3**	**Course Code**	**23CT1301**
Roll No	**13**	**Name of Student**	**Juhi Parwani**

Practical Number	1(A)
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	To develop a TypeScript-based To-Do List application that allows users to add tasks, mark tasks as completed, and delete tasks using type-safe programming concepts.
Problem Definition	To design and implement a To-Do List application using TypeScript that manages tasks through typed data structures and functions. The application should allow users to add new tasks, mark existing tasks as completed or incomplete, delete tasks, and display task information through a web interface.
Theory (100 words)	TypeScript is a superset of JavaScript that provides static typing and interfaces to improve code reliability and maintainability. In this practical, TypeScript is used to define a `Task` interface containing an ID, title, and completion status. Typed functions are implemented to add, complete, retrieve, and delete tasks. Arrays and methods such as `map()` and `filter()` are used to update and manage task data. TypeScript code is compiled into JavaScript, which can execute in a web browser. This practical demonstrates how type safety helps organize application logic and reduces programming errors while developing a simple task management application.

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Procedure and Execution

(100 Words)

Step for Implementation:

Create a folder named `Practical_8A`.
Create the TypeScript files `types.ts`, `todoLogic.ts`, and `main.ts`.
Define a `Task` interface with `id`, `title`, and `completed` properties.
Implement functions to add, complete, retrieve, and delete tasks.
Write test cases in `main.ts` to demonstrate task operations.
Compile the TypeScript files into JavaScript using the TypeScript compiler.
Execute the compiled JavaScript using Node.js and verify the output.
Create `index.html` and `web.ts` to provide a browser-based interface.
Compile `web.ts` into `app.js`.
Run a local web server and test adding, completing, and deleting tasks in the browser.

Code:

types.ts

```
export interface Task {
  id: number;
  title: string;
  completed: boolean;
}
```

todoLogic.ts

```
import type { Task } from "./types";

let tasks: Task[] = [];
let nextId = 1;

export function addTask(title: string): Task {
  const task: Task = {
    id: nextId++,
    title,
    completed: false,
  };

  tasks.push(task);
  return task;
}
```

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```
export function completeTask(id: number): void {
  tasks = tasks.map((task) =>
    task.id === id ? { ...task, completed: true } : task
  );
}

export function deleteTask(id: number): void {
  tasks = tasks.filter((task) => task.id !== id);
}

export function getTasks(): Task[] {
  return tasks;
}
```

main.ts

```
import {
  addTask,
  completeTask,
  deleteTask,
  getTasks,
} from "./todoLogic";

console.log("Adding tasks...");

const task1 = addTask("Complete Practical 8");
addTask("Study TypeScript");

console.log("All tasks:", getTasks());

completeTask(task1.id);
console.log("After completing task:", getTasks());

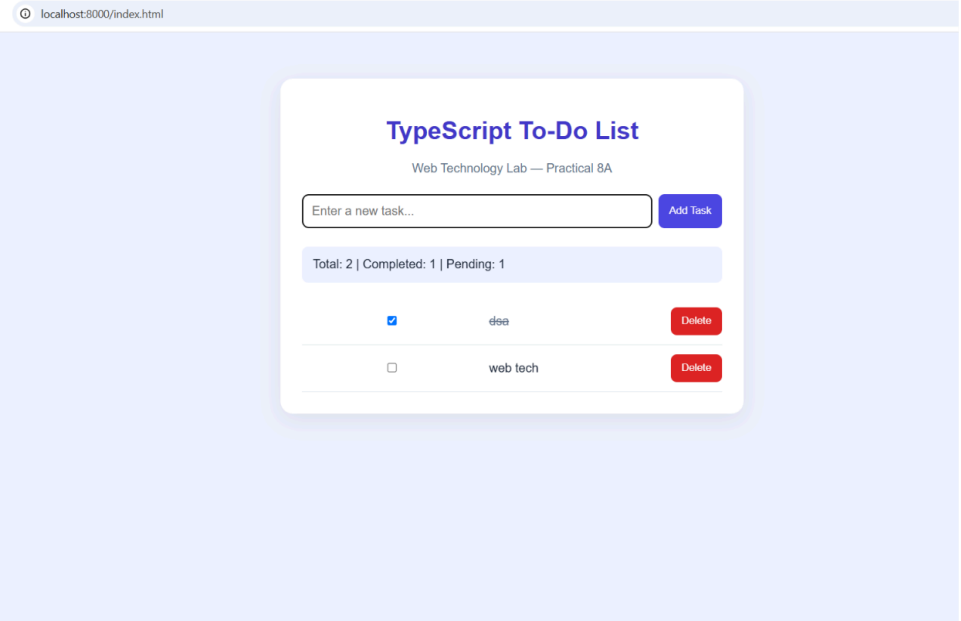
deleteTask(task1.id);
console.log("After deleting task:", getTasks());
```

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	Output: 
Output Analysis	The application successfully demonstrates task management using TypeScript. The `Task` interface ensures that each task follows a defined structure. The `addTask()` function creates tasks with unique IDs, `completeTask()` updates the completion status, and `deleteTask()` removes a task from the array. The browser interface allows users to perform these operations interactively and displays updated task counts. The results confirm that the application works as intended.
Link of student Github profile where lab assignment has been uploaded	**https://github.com/juhiparwa-1507/Web-Technology-Lab.git**
Conclusion	Thus, the TypeScript To-Do List application was successfully implemented and tested. The practical demonstrated interfaces, type annotations, arrays, functions, and array methods

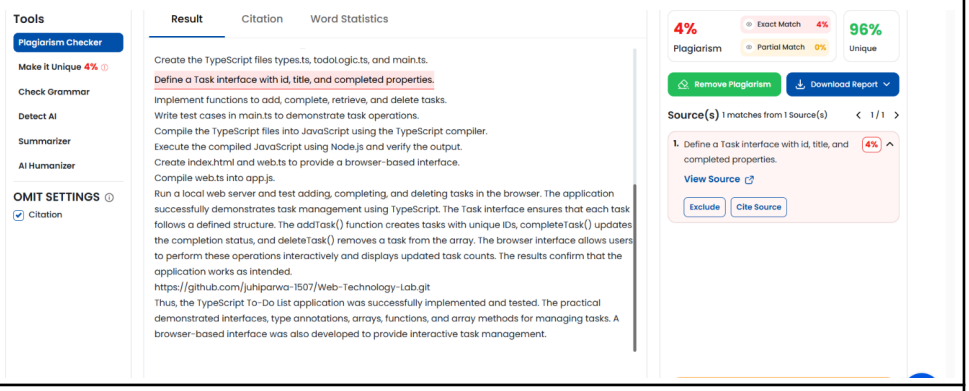
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	for managing tasks. A browser-based interface was also developed to provide interactive task management.
Plag Report (Similarity index < 12%)	
Date	**14-10-26**